Structuring Interviews with Experts During Knowledge Elicitation

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We have developed interviewing techniques for knowledge elicitation based on (1) relevant research and techniques from the social sciences, (2) the nature of expertise, (3) a desire to assist a knowledge engineer to avoid reductive bias, one of the pitfalls associated with the acquisition of highly complex concepts, and (4) the desirability of de-coupling elicitation from implementation. The approach consists of four phases with guidelines for questions suited to each stage. First, the descriptive elicitation stage is intended to reveal the important entities and concepts in the domain as reflected in the terms and specialized language used by the expert. A second stage, structured expansion, is designed to probe the relationships between the domain concepts and the organization of the expert's knowledge using the terminology uncovered in the previous stage. The scripting phase relies on the declarative knowledge found through the two previous stages to discover procedural knowledge, and the final component, validation, is important throughout the process of knowledge elicitation to ensure that the knowledge being elicited is correct and adequate to enable a system in which it is implemented to solve the class of problems with which it is concerned. © 1993 John Wiley & Sons, Inc.

I. INTRODUCTION

Recently, Shaw and Woodward[1] proposed a framework in which knowledge acquisition methods and issues can be addressed in a systematic fashion. Their proposals are a reaction to their characterization of the field of knowledge acquisition as a methodological "patchwork," resulting from the importation and adaptation of methods from other disciplines. We have attempted to use Shaw and Woodward's framework as an organizing scheme in the present article to report our own work in developing structured interviews for the elicitation phase of knowledge acquisition. At the outset, we would like to clarify the use of the terms knowledge "acquisition" and knowledge "elicitation," following the distinction made by Motta *et al.*[2]. Knowledge "acquisition" will refer to the broader activity of gathering information from a variety of sources, one of which is a domain expert, and interpreting and organizing it. The activities

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involving direct interactions with an expert will be referred as knowledge "elicitation." We make this distinction because the work reported herein is concerned primarily with the latter activity.

The concerns expressed by Shaw and Woodward[1] about the field of knowledge acquisition, generally, have been expressed by Forsythe and Buchanan[3] about interviewing techniques for knowledge elicitation, more specifically. They point out that "while recent publications have come closer to addressing some of the practicalities of interviewing for knowledge acquisition, they still suffer from the general tendency to name methods without providing much information on how to apply them. . . . In short, we know a good deal about how to represent and manipulate knowledge in expert systems, but have very little systematic understanding of how best to gather that knowledge in the first place" (pp. 1–2). They then proceed to report some of their work designed to address this shortcoming. Whereas the issues addressed by Forsythe and Buchanan are directed at the social context and communication issues surrounding an interview with a domain expert, the work reported here is concerned with particular kinds of questions asked during an interview and with guidelines for when and how to ask them.

II. GUIDING PRINCIPLES AND ASSUMPTIONS

Although our work is centered around interviewing techniques, we do not mean to imply that we consider interviews to be adequate and sufficient for meeting all the needs of knowledge elicitation. Rather, our goal is to provide guidelines for making interviews with an expert as productive as possible. Also, as will become apparent, our conception of what constitutes interviewing is reasonably broad. Because our approach involves the importation and adaptation of methods from several disciplines, we attempt to provide theoretical bases for them grounded in those disciplines. Also, in keeping with the recommendations of Shaw and Woodward, we discuss the working assumptions and principles that have guided our work before describing specific aspects of our techniques.

A. The Nature of Expertise

It is not uncommon for a report on knowledge acquisition to begin with a statement to the effect that knowledge acquisition continues to be the major bottleneck in the process of building knowledge-based systems. As suggested by LaFrance,[4] one reason for this is that developers continue to underestimate the complexity of expertise. Indeed, this is consistent with our own experience and assessment of the knowledge acquisition literature. Thus one of the guiding principles underlying our approach to knowledge elicitation has been to consider the nature of expertise as it has been characterized by researchers in the discipline of cognitive psychology. As a result, we will briefly review some of the characteristics of expert knowledge that we feel should be considered by

those involved in knowledge acquisition. For a more thorough treatment of this issue consult LaFrance.[4]

Four aspects of expertise which we find particularly relevant to knowledge elicitation are the organization of knowledge, its influence on the representation of problems, characteristic problem-solving strategies, and the tacit nature of the knowledge. In addition to being sensitive to these four aspects of expertise, our methodology attempts to avoid "reductive bias," one of the pitfalls of advanced learning in complex domains.

1. Knowledge Organization

Experts have a great deal of domain-specific knowledge, compared to novices. However, more than quantity of knowledge, it is the qualitative differences that have implications for knowledge elicitation.[4] At a macro level, a variety of studies (e.g., Chi *et al.*[5]) have found expert knowledge to have a generally hierarchical organization (i.e., categories with subcategories), although many other types of relationships are also present, depending on the particular domain. All of this domain-specific knowledge is tightly interconnected with many associations among items. By contrast, novices do not yet have their domain-specific knowledge nearly as well organized or tightly integrated.

At the micro level, expert knowledge is stored as organized "chunks" of frequently occurring patterns or schemas.[4] In the case of chess, for example, Simon and Gilmartin[6] have estimated that grand masters have acquired on the order of 50,000 different patterns or chunks of chess information, each of which is associated with one or more strategies used in play. An important implication of this chunking or schema formation is that it enables an expert to quickly recognize patterns in the information encountered during problem solving. Part of schema formation is the association of functional characteristics and action sequences relevant to that particular pattern.

While expert knowledge has abstract dimensions, LaFrance[4] points out that it is important to recognize that experience is grounded in specific situations or "cases" that have been encountered in the past. While abstractions obviously arise from those specific cases, we must not lose sight of the fact that they remain an important part of the memory organization of an expert and provide an important basis for problem solving.[7]

2. Problem Representation

Differences in amount and organization of knowledge naturally lead to differences in the way novices and experts represent problem information as they analyze it during problem solving. For experts, representation generally takes the form of underlying principles and abstractions.[5,8] This is related to the notion of schemas discussed above. In contrast, novices tend to focus on superficial characteristics of a problem and objects described in the problem statement. In computer programming, for example, problem representation of

experts takes the form of language-independent, abstract constructs, whereas novices conceptualize problems in terms of commands and control structures specific to the programming language in which they are working.[9]

3. *Problem-solving Strategies*

Some of the characteristic ways in which experts approach problems carry across domains, while others are domain specific. One general approach is a working-forward strategy, which means moving from the information given in the problem to the goal via operations that the expert knows will work. In contrast, novices tend to begin at the goal, noting information that is missing. They then work backward one step at a time, determining at each point what is needed to solve the immediately succeeding step. Once all missing information is found, the steps can be reversed to solve the problem. This is particularly clear, for example, in the way people approach geometry proofs[10] and physics problems.[11]

Researchers have also documented a number of domain-specific problem-solving strategies. For example, in medical diagnosis experts develop the ability to entertain several competing hypotheses about the disease simultaneously, whereas novices tend to consider only one at a time.[12] This seems directly related to the larger degree of integration found in the experts' knowledge about diseases and symptoms and their relationships. Another example of domain-specific problem-solving strategies comes from computer programming. In this case experts take a top-down approach to program development, working at a global level and developing the program in a breadth-first manner. In contrast, novices tend to focus on low-level details in what is referred to as a depth-first strategy.[9]

4. *Tacit Knowledge*

It is a common observation that much of an expert's problem-solving knowledge has become automatized or tacit through extensive use, and there has been considerable debate about its accessibility, even to the expert.[13,14] Anderson's[15] account of the development of automatized knowledge describes this difficulty. According to Anderson, declarative knowledge includes the conceptual entities (both concrete and abstract) in a particular domain and their various relationships. Procedural knowledge, on the other hand, is comprised of the knowledge required for solving problems using relevant domain concepts and relationships. Another way to describe this distinction is a knowledge of "what" vs a knowledge of "how."

In early stages of skill learning an individual consciously considers various items of declarative knowledge during problem solving. In well learned tasks, much declarative knowledge is no longer consciously available during problem solving, although this does not mean that the declarative knowledge has been "forgotten." It does mean that it may be difficult for an expert to articulate it to a knowledge engineer, especially when asked to do so directly.

B. Reductive Bias

Because a knowledge engineer is often a relative novice in the domain of knowledge being elicited, there exists a potential for errors stemming from what has been termed "reductive bias." This is a problem that arises in situations involving the learning of complex bodies of knowledge, such as medicine. For example, Feltovich *et al.*[16] have researched the basis for significant, pervasive misconceptions that occur in medical students as they study to become specialized practitioners. They describe a particular case in which a serious misconception about the basis of congestive heart failure was found to exist not only in the minds of medical students, but also in the minds of some medical practitioners, and in some medical textbooks.

Feltovich *et al.*[16] outline characteristics of advanced "knowledge acquisition" that make unusual demands on cognition which tend to promote reductive (or simplification) biases. These, in turn, lead to misconceptions on the part of the student. Two examples of reductive bias are analogy bias and common connotation bias. The former results from the inappropriate application of an analogy that is too simple, or the inappropriate extension of attributes (e.g., if A and B are alike on attribute X, they are also alike on attributes Y and Z) Common connotation bias occurs when everyday terms are used in technical ways, but not understood (by the student) for their technical meaning.

In attempt to minimize the potential effects of reductive bias, we have imported techniques from the disciplines of anthropology and ethnography because researchers in those disciplines face a similar problem when they are attempting to elicit cultural information from informants (or cultural "experts") in unfamiliar societies. Spradley[17] makes a convincing case for the importance of language in this exercise. He notes that this can be particularly critical when the members of the culture appear to speak the same language as the researcher because important semantic differences exist that are not readily apparent. This is essentially the "common connotation bias" of Feltovich *et al.*[16] referred to above.

Spradley[17] describes another potential source of bias, which he terms "translation competence." This arises when informants translate the reality of their culture in order to explain it to an outsider. The problem is that the more an informant translates for the convenience of an investigator, the more that informant's cultural reality becomes oversimplified and distorted.

In an effort to avoid translation bias, Spradley[17] developed an approach to questioning that makes minimal assumptions about informants' knowledge and uses information they provide as the basis for further questioning. The researcher first uses very general probing techniques to persuade the informant to talk freely about his or her domain in a global sense. The informant's language is recorded and examined for category labels and other linguistic cues. This information is then used to cue the informant for additional information. Throughout the process it is necessary to verify that the researcher's emerging picture of the domain accurately reflects the informant's cultural expertise. We have adopted this general framework in which to develop a knowledge elicitation methodology.